Problem statement & objectives

Formula 67 — Project proposal (v4.0)
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Motivation: Many student racing prototypes implement movement and scoring only. Formula 67 extends a classic top-down racer with a telemetry layer so driving behaviour can be inspected after each session using tables and charts, mirroring real motorsports analytics at a small scale.

User story: A player selects a difficulty, completes a three-lap race against AI opponents on a fixed circuit, and receives immediate in-game feedback (HUD + results). Afterward, CSV logs capture speed samples, steering events, lap segments, collisions, and nitro bursts; a matplotlib script aggregates those logs into a single telemetry figure saved under reports/.

Scope, data pipeline, plan

Technical scope: Python 3, pygame for rendering and interaction, mask collision for track limits and finish line, AABB tests for obstacles and nitro pads, and a RaceManager coordinating laps and timing. StatsLogger buffers structured rows and exports UTF-8 CSV files; Leaderboard persists the top ten lap times.

Data & visualization: At least five distinct features (speed time series, steering events, lap times, collisions, nitro-related metrics) are accumulated across races. visualize.py reads stats/*.csv and builds a multi-panel report containing tables, a line chart, histogram, scatter, and pie chart where applicable.

Development plan: (1) core OOP structure — Track, Car, RaceManager; (2) obstacle and nitro systems; (3) AI path followers; (4) StatsLogger integration; (5) matplotlib reporting; (6) polish, HUD, menus, documentation, and submission packaging.

Success criteria: Stable three-lap loop; demonstrable inheritance (PlayerCar extends Car); clear module boundaries; reproducible CSV export; generated visualization artefact; README and DESCRIPTION matching course outlines; LICENSE and attribution for external art.